FOURTH PARTNER ENERGY

CASE STUDY · PRODUCT & OPERATIONS INTERNSHIP

# Kit-to-Site

*Rethinking how material reaches the rooftop*



PREPARED FOR
Gentyal Abhinav, IIT Kharagpur

ISSUED
October 2026

TIME ALLOWED
5 days

# About this case

Thank you for your interest in the Product & Operations Internship at Fourth Partner Energy. This case is based on a decision our operations team is working through now, and there is no single right answer. We want to see how you approach an unfamiliar problem with imperfect data, and what you recommend.

## THE SITUATION

Fourth Partner Energy (FPEL) designs, builds and operates rooftop solar plants for commercial and industrial customers across India. Between July and September 2026 it delivered 40 rooftop projects, supplied from three regional stores.

**40**
rooftop projects, July to September 2026

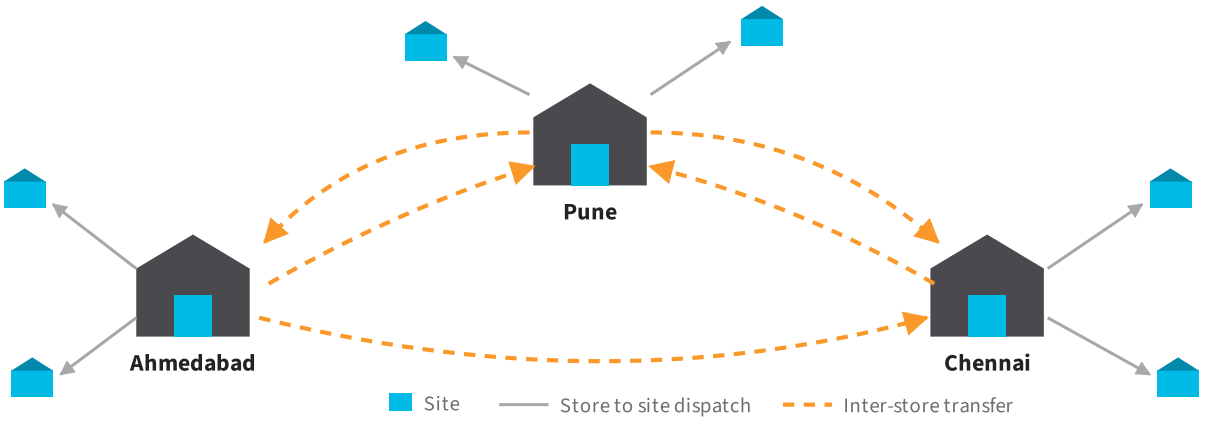
**~14**
MWp of capacity installed or in progress

**3**
regional stores: Pune, Ahmedabad, Chennai

**5**
installation stages on every rooftop

Two issues keep coming up. Installation crews lose working days waiting for material, and stores often ship stock to each other to cover gaps. Management is considering a larger central warehouse and wants to understand how material should flow before committing to one.

**EXHIBIT 1** How material moves today




Each store supplies the sites in its region. When a site runs short, stock moves between stores.

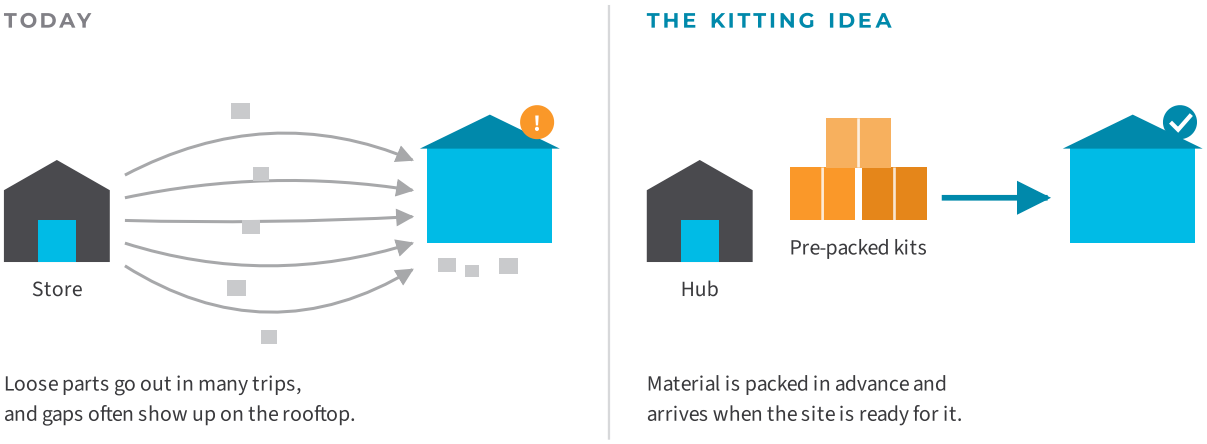
**IN SCOPE**
Material flow from supplier to rooftop, kitting, warehouse capacity and purchasing rules for three commodity-linked items.

**OUT OF SCOPE**
Plant engineering design, customer pricing, installation methods and supplier selection.

# Could kitting work for rooftop solar?

Elevator manufacturers such as Otis send pre-engineered kits to installation sites, so crews get what each step of the job needs at the time they need it. FPEL wants to know whether the same approach would work for rooftop solar, and what it would mean for the warehouse and for purchasing.

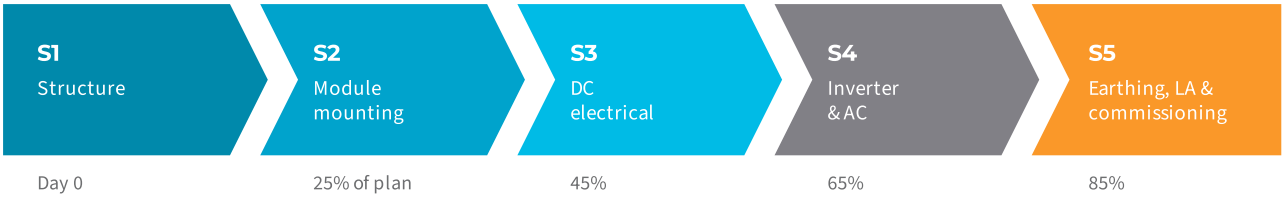
## EXHIBIT 2 Loose parts today, and the kitting concept




The concept only. How FPEL should define, pack and deliver kits is for you to work out.

## EXHIBIT 3 How a rooftop is built

Every project goes through the same five installation stages, each starting at a fixed point in the planned schedule.



# Four questions to answer

THE KEY QUESTION

**How should FPEL move material from suppliers to rooftops, and what does that mean for the warehouse it is about to commit to?**

## 01

### Diagnose

What is causing crews to wait on site, and stock to move between stores?

- Quantify the time and cost involved
- Separate root causes from symptoms

## 02

### Design

What should a kitting system for FPEL look like?

- What is kitted, and what is not
- How kits are structured, and why
- Which items could go straight to site
- Where kitting happens and how customer-specific items are handled
- Show the kit list for one project from the data

## 03

### Plan purchasing and capacity

For fabricated GI structure, 4 sq mm DC cable and aluminium AC cable:

- Propose practical rules for when to buy and how much to hold
- Estimate the floor area FPEL actually needs
- Say whether a larger warehouse makes sense

## 04

### Pilot

How would you test your design over 60 days?

- Which region and projects to include
- The KPIs you would track, and how
- The main risks, and what would make you change course

# What colleagues told us

We spoke with four colleagues while preparing this case. Their views are genuine but partial, and they don't always agree. Weigh them against the data.

*"Every project loses about a week waiting for material, and my crews can't plan around shortages. If we sent the full material to site on day one, the problem would go away."*

R **Rakesh**
Head of Projects

*"Pune is at capacity, and modules take up more than half the floor. Requests come in at the last minute, so we keep shipping stock between stores. We need the bigger warehouse soon."*

M **Meena**
Stores In-charge, Pune

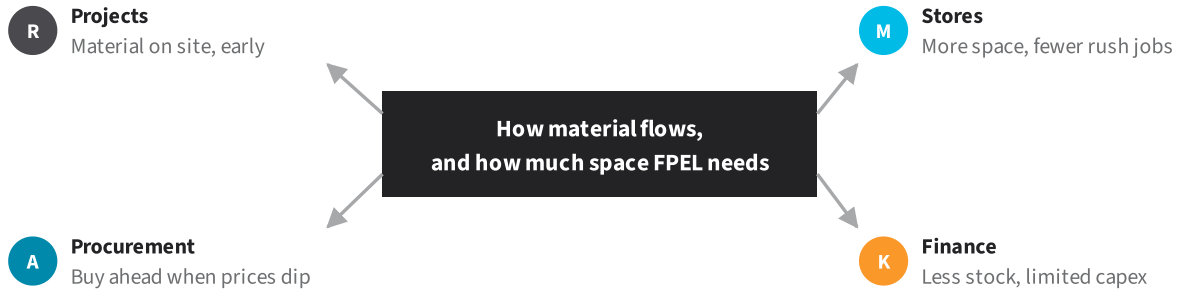
*"Copper is close to its six-month low and well below its yearly average. I'd like approval to buy six months of DC cable now, and to lock steel prices for the next quarter."*

A **Arjun**
Head of Procurement

*"We are carrying about Rs 25 crore of stock, close to two months of material. I'd like inventory days down by a third by March, and warehouse fit-out capex is capped at Rs 50 lakh."*

K **Kavita**
Finance Controller

## EXHIBIT 4 Four priorities pulling in different directions




Each colleague is responding to a real pressure on the business. Part of the task is deciding which concerns your recommendation deals with first, and saying why.

# Kit-to-Site_Data_Pack.xlsx

The workbook that comes with this case has ten sheets of operating data. The data is illustrative but modelled on real operations, and like most operating data it was pulled together from several systems.

| 40 | 1,522 | 76 | 71 | 105 |
|---|---|---|---|---|
| projects | material issue lines | site delay entries | inter-store transfers | pipeline projects |

EXHIBIT 5 **What is in the workbook**

**README**
Sheet guide, units and installation stages

**Projects**
40 projects: size, roof type, inverter brand, site conditions, dates

**BOM_Master**
Quantity rules, install stage, price, floor space, lead time per item

**Issue_Log**
Every issue from store to site, with one reference per vehicle trip

**Delay_Log**
Site delays with days lost and site engineers' remarks

**Transfer_Log**
Stock moved between stores, with freight cost

**Warehouse_Stock**
Stock on hand at each store on 30-09-2026, with floor space used

**Pipeline**
Projects expected to start October 2026 to March 2027, by sales stage

**Price_History**
12 months of commodity indices and FPEL purchase prices

**Assumptions**
Store details, cost assumptions and pipeline conversion history

> Treat this as you would real operating data. Where something is missing or unclear, make a reasonable assumption, state it and move on. All figures are in Indian Rupees and metric units.

# Deliverables and timeline

### Recommendation deck

PDF · UP TO 12 SLIDES

Start with an executive summary of your recommendation. Appendix slides are welcome.

### Supporting model

EXCEL

Your analysis, laid out so a reviewer can trace each output back to its inputs.

### Decision log

1 PAGE

The five decisions that shaped your recommendation, the alternatives you considered, and what would make you revisit each one.

### Note on AI use

HALF A PAGE

The tools you used and what you used them for. Include at least one place where you chose to go against the AI's suggestion, and explain why.

BONUS

### Material buy-or-hold forecaster

Build a working tool for FPEL's main commodity-linked materials (fabricated GI structure, DC cable and AC cable) that:

- takes the latest commodity prices and the project pipeline as inputs
- forecasts prices and material demand for the next six months
- recommends whether to buy now or hold for each material, with a suggested quantity and timing, and shows the reasoning behind it

Use any platform, including spreadsheets, no-code tools and AI app builders. Submit the working tool, as a file or a live link, with a short note on how to run it.

## TIMELINE Five days from receipt




**Day 0**
Case received

**By day 2**
Send any clarifying questions (one round, by email)

**Day 5, 18:00 IST**
Submit all deliverables

To submit, reply to the email this case came with. We expect the case to take about 20 to 25 hours of focused work over the five days.

# How we will look at your work

We read each submission as a whole. Broadly, we look for:

| Structured problem solving | Rigour with data | Practical design | Judgment | Clear communication |
|---|---|---|---|---|
| A clear line from question to answer | Careful reading and sound analysis | Ideas that would work on a real rooftop | Trade-offs made and explained | A recommendation a leader can act on |

## Start from the question

Fewer slides that answer the question well are worth more than many that describe the data.

## State your assumptions

You will have to make judgement calls. Say what you assumed and why, so a reader can follow your reasoning and test it.

## Using AI tools

You are welcome to use AI tools. We are most interested in your own reasoning, so make sure you fully understand, and can stand behind, every number and recommendation you submit.

## Questions

If anything in the brief is unclear, send your questions in a single email by the end of day 2. We can clarify the brief but will not provide additional data.

## Confidentiality

This case and its data are shared for this assessment only. Please do not share, publish or reuse them outside the selection process.

## Pace yourself

Plan for about 20 to 25 hours. If time runs short, put a clear recommendation ahead of completeness.

## We look forward to reading your work.

Good luck.